

Backup Infrastructure — Target-State Design & Staged Execution Roadmap

Keep T-Net licence + off-site; redesign, reclaim space, optimise, re-replicate

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Scope. The environment is recovered and restore-proven. This is the optimisation plan: clean up the accumulated mess, reclaim wasted space, and settle a durable design — keeping T-Net's Veeam licence and off-site Cloud Connect, but with KOR owning the on-prem design and operations. Every step is tagged for who does it.

1. Where we are (verified 2026-07-23)

- **Veeam 12.3.2** on KOR-BK01 (Server 2019 VM). Enterprise Plus (T-Net rental).
- **KOR-FS01** (21 TB) protected: dedicated **forward-incremental** job → Synology, mount server **KOR-APP01** (Server 2025). **File-level restore of the deduplicated volume works.**
- **Small VMs** (APP01, DC01, RDS01) + vCenter: nightly reverse-incremental → E:.
- **Off-site:** Cloud Connect at `remote.t-net.ca` — holds the OLD chain (frozen at July 7). The new FS01 chain is not yet off-site.
- **All three copies restore-tested** (E: pre-dedup, Cloud, Synology post-dedup). Email alerting live.

The problems this design fixes

Issue	Impact
E: (25.4 TB) over-committed — ~23 TB is FS01's old reverse-incremental chain	Chronic low-space warnings; no headroom; caused the original failure
Reverse incremental on large VMs	3× write penalty; ~29 MB/s vs 126 MB/s forward; couldn't complete
Retired BMZ VMs still registered on host .16 (FS007 10.5 TB, DC005 9.1 TB, +APP002, RDS001)	~20 TB of production SAN wasted
Duplicate <code>Kor-RDS01</code> / <code>Kor-RDS01_1</code> ; <code>ManualNew</code> job of unknown origin; orphaned catalogue records	Confusion, wasted space, risk
FS01 not replicating off-site since the rebuild	No current off-site copy of the file server

2. Target architecture

- Production** (vSphere, UC3200 SAN) → **Veeam BK01** (12.3) →
- **Primary backup** → **Synology (F:)**, forward incremental + weekly synthetic full, mount server APP01
 - **Off-site copy** → **Cloud Connect (T-Net Toronto)** via WAN-accelerated Backup Copy
 - **E: repurposed** as a second local copy target (optional) once its old chain is retired

Core design decisions:

- **Forward incremental + weekly synthetic full everywhere.** Proven faster and smaller here; protects against chain corruption; plays well with modern storage. (This is also what T-Net's "synthetic fulls with incremental" actually describes.)
- **Synology is primary.** It's where FS01 already works, with the APP01 mount server that makes deduplicated restore possible.
- **Back up only live VMs** (~5). Retired VMs are removed from jobs and from production.
- **Right-size, don't trim history.** Reclaim space by removing dead data, not by shortening retention on live systems.

3. Space reclamation — the big wins

Target	Reclaims	Where	Owner
Retire BMZ-HO-FS007, DC005, APP002, RDS001 (verify FS007 first — old file server, may hold only copy)	~20 TB	Production SAN (UC3200)	KOR SOLO
Retire the old FS01 reverse-incremental chain on E: (only after off-site + all restores confirmed)	~23 TB	E: repository	COORDINATE
Delete orphaned Dec-2025 .vrbl files	~44 GB	E: repository	KOR SOLO
Resolve Kor-RDS01 / Kor-RDS01_1 duplicate	TBD	SAN + backups	KOR SOLO
Clarify / retire ManualNew (unknown origin, holds a Mar-2026 FS01 full on Synology)	up to ~few TB	Synology	CONFIRM ORIGIN

Net: ~20 TB back on production immediately (dead VMs), and up to ~23 TB on E: once the old chain is safely retired — turning a chronically-full repository into a right-sized one.

4. Off-site re-replication — getting FS01 to Toronto

Method: a WAN-accelerated Backup Copy job from the FS01 Synology chain → the Cloud Connect repository, throttled during business hours.

Seed estimate (office upload measured at 182 Mbps, 5.8 TB to move):

Approach	Initial seed	Ongoing
Full bandwidth, off-hours (nights + weekends)	~3–4 days	Daily incrementals ~20–40 min; WAN accel shrinks further
Throttled ~50% (office keeps bandwidth in work hours)	~6–7 days	

Resumable — survives interruptions. **Ask T-Net:** does the Toronto provider support a shipped seed-drive import? If yes, the initial ~week collapses to a courier trip; if no, the wire is entirely workable.

5. Instant VM Recovery — capability & RTO

- **Supported and available now** — the APP01 mount server (vPower NFS) is exactly what IVMR uses. Nothing to "turn on."
- **Most valuable for FS01:** if the 21 TB file server dies, boot it from backup in minutes and run degraded while restoring — vs a multi-hour full restore.
- **Constraint:** IVMR's write cache lands on the mount server's C:, and APP01's C: is tight (~43 GB). Fine for a quick small-VM boot; limiting for running FS01 from backup for long. **If FS01 RTO matters, add a dedicated cache disk to APP01.**
- **Action:** validate with a test IVMR of a small VM (same discipline as the restore tests).

6. Staged execution roadmap

#	Step	Owner	Depends on
0	Veeam 12.3 upgrade · APP01 mount server · deduped FLR proven · all three copies restore-tested · alerting live · snapshot cleaned	DONE	—
1	Delete orphaned Dec-2025 <code>.vrb</code> files (safe filesystem reclaim)	KOR SOLO	—
2	Verify BMZ-HO-FS007 / DC005 / APP002 / RDS001 hold nothing needed; archive if any doubt; retire from vSphere → reclaim ~20 TB SAN	KOR SOLO	Data check
3	Resolve RDS01 / RDS01_1 duplicate (confirm live disks, clean the stale one)	KOR SOLO	—
4	Confirm <code>ManualNew</code> origin; retire or repurpose	CONFIRM W/ T-NET	—
5	Convert small-VM jobs (Kor-VMs-New, Vcenter) to forward incremental + weekly synthetic full	KOR SOLO	—
6	Build WAN-accelerated Backup Copy: FS01 Synology → Cloud Connect; throttle, seed over ~1 week	WITH T-NET	T-Net endpoint + WAN accelerator
7	Validate IVMR with a small-VM test; decide on APP01 cache disk for FS01 RTO	KOR SOLO	—
8	Once FS01 off-site + all restores confirmed: retire the old E: reverse-incremental chain → reclaim ~23 TB; repurpose E: as second local copy (optional)	COORDINATE	Steps 6, and restore confirmations
9	Optional optimisation: ReFS-64K on the backup volume for fast-clone (faster synthetic fulls, less space)	KOR SOLO	Volume reformat plan
10	Establish restore-test cadence (quarterly file-level, annual full-VM/IVMR)	KOR SOLO	—

7. What we need from T-Net

- **Off-site re-seed for FS01** — enable/confirm the WAN-accelerated Backup Copy to the Cloud Connect endpoint; confirm whether a shipped seed-drive import is available at the Toronto provider. *(This is the priority — it closes the off-site gap.)*
- **Licence continuity** — confirm the Enterprise Plus rental covers our continued use through these changes (and the upgrade already applied).
- **Confirm** `ManualNew` — is it theirs, and what is it for?
- **Certificate ownership** — logged renewal ownership ahead of 2028-07-14, in their ticketing system.
- **Monitoring/alerting SLA** — direct job-failure alerts to KOR + defined response, so an extended undetected failure can't recur.

Internal design document, KOR Structural Operations, 2026-07-23. Seed estimates based on a measured 182 Mbps office upload. Steps tagged "KOR solo" are executable without provider involvement; "Needs T-Net" items require coordination with the licence/off-site provider. This document doubles as the execution roadmap and the scoped ask to T-Net.